Movie Box Office Success Prediction using Machine Learning

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***Abstract* — Predicting societal reactions to a new product in terms of perceived popularity and rate of adoption has become a field of data analysis. The film industry is a multibillion-dollar industry, and there is a vast amount of information about movies on the internet. This research proposes a machine learning-based solution for the film investing business. The method analyses previous data from several sources to forecast a movie's success rate. K Nearest Neighbors (KNN) is a method for predicting the box office success of a film.**

Keywords—movie industry, Knearest neighbors algorithm.

1.INTRODUCTION

The film industry is a massive investment sector, but vast company sectors are more complex, making it harder to decide where to put your money. Because the film industry is expanding at an alarming rate, there is already a significant amount of data available on the internet, making data analysis a fascinating field. The measure of a film's success is subjective; some videos are deemed successful based on their worldwide gross revenue, while others may not perform well commercially but receive positive critical reviews and popularity. In this post, we'll look at a successful box office film. In comparison to traditional media, there has been a significant shift in the social media sector in recent years. We need a powerful approach to quickly find facts that can be utilised to frame a certain subject. The rapid increase in the quantity of films released in recent years necessitates a few decades of movie forecasting. In other words, the goal is to create learning algorithms that can be used by machines without the need for human interaction. The goal of this research is to present a machine learning approach that provides the most accurate outcome for movie success prediction.The factors can determine the success of a movie:

- The popularity of film content.
- The present popularity of the film's actors.
- The strength of the cast and the film's release schedule.
- Awareness of the film based on its adaptation of a popular book or report

11.LITERATURE REVIEW

M. T. Lash and K. Zhao[1] developed a decision support system that predicts the profitability of a film using machine learning approaches, text mining, and social network analysis.

Sivasantoshreddy et al.[2] attempted to anticipate a movie's box office gross opening. Their study focused on the use of Twitter data for hype analysis. The fundamental rationale behind hype analysis was to determine a film's performance based on its opening weekend box office and how much buzz it generated before its release.

Lydia [3] used news analysis and quantitative news data to try to increase the gross profit of the film prediction.

Jonas et al. [4] employed sentiment and social media analysis to predict, and their hypothesis was based on IMDb's Oscar Buzz sub-intensity forum's and positivity analysis. They had viewed cinema reviewers as a source of influence and foresight. They employed a collection of terms that produced poor results when particular words were used for negative purposes. There was no category award; instead, only the awards for best picture, director, actor/actress, and supporting actor/actress were considered.

111.MOTIVATION

Nowadays, audiovisual input and output devices are widely used. People are watching movies in greater numbers than ever before. As a result, the film business is arguably one of modern society's most significant sectors. Every film has the power to reflect society and change people's minds. In many ways, a good film may entertain, educate, and inspire its audience. Consider the impact that music has on people. Each film is set in and developed within a specific culture. Movies have become an inseparable part of our lives.

V1.METHODOLOGY

The purpose of this paper is to find a Machine Learning model to identify the box office success of a movie using the givendataset with accuracy. The model should be able to determine correctly as the movie's box office success.The dataset is divided into two training and testing datasets.

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The model should be trained with more data to get accuracy in learning. The result from this model can be used to create a movie's classification. The knn algorithm is used to determine the box office success of a movie. The structure is:

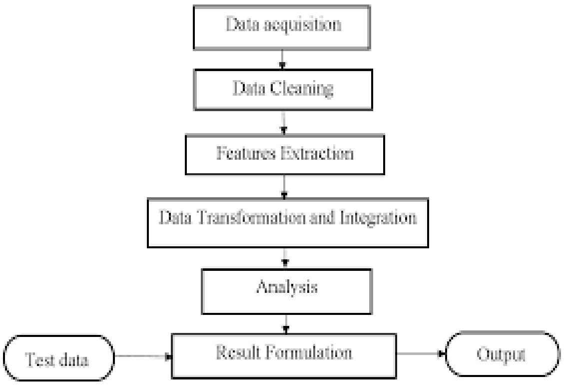

Figure 1: Movie Success Prediction Methodology

The steps that require to be followed are:

1. Data Collection
2. Data Pre-processing
3. Model Building
4. Analyzing
5. Result

A. **Data Collection:** User have a dataset for movie box office success, the attributes of this dataset is used to train the model. There 839 rows and 6 columns in dataset. The columns are headings. The heading denotes the movie's headings. This is to find the movie is success or not.

B. **Data Pre-processing:** This is the step that is done in every machine learning method. This process includes data cleaning, data transformation and data reduction. This all are done to make the data more effective. The data can be processed to make our model more accurate. So that the classification will be correct.

1.Cleaning Data – The data is cleaned at first. That is here users are removing the unwanted entries. The noisy data can also be removed from the dataset.

2.Removing Stop words – The stop words are removed from the datasets. These are the most frequently appearing that don't have much importance in the text. To achieve more accuracy to the model this can be done.

3.Splitting of Data – After cleaning the data the data is then split into training and testing datasets. After this the data chosen for training is used to train our model.

C.Machine Learning: This is used to determine the movie success. The machine learning algorithms are used to determine the movie's success, that will check the given movie. There are different classifiers present to determine the movie's success. The classifier accuracy relies on how it is trained. A classifier should need high accuracy.

KNN- K –Nearest Neighbor is a supervised learning machine learning method that is one of the most basic. The K-NN algorithm assumes that the new case/new data and the existing cases are comparable, and it places the new example in the category that is most similar to the existing categories. This algorithm saves all available data and rates fresh data points according to how similar they are. This means that when fresh data comes, the knn algorithm can quickly place it into a well-suited category.

The following stages can be used to explain the knn algorithm.
Step 1: Determine the number of neighbours (K).

Step 2: Determine the distance between each of the K neighbours.

Step 3: Count the number of neighbours who are closest to you.

Step 4: Count how many data points there are.

Step 5: Assign the new data to the appropriate field.

Step 6: The model is complete.

V.BUILD MODEL

The model building is the main step in the movie box office success. While building the model user use the algorithms. The steps involved are:

1. Import the packages that are necessary.

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import confusion_matrix
from sklearn.metrics import accuracy_score
from sklearn.metrics import precision_score
from sklearn.metrics import recall_score
from sklearn.metrics import roc_curve, auc
import matplotlib.pyplot as plt
import random
```

2.To read the dataset.

```
data=pd.read_csv("Dataset.csv")
data=data.dropna(axis=0, how='any')
```

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3.To analyzing the data.

```
X = data[data.columns[6:32]]
Y=data.iloc[:,-1]
```

4.To split dataset into test dataset and training dataset.

```
X_train,X_test,Y_train,Y_test=train_test_split(X,Y,test_size=0.25,random_state=0)
scaler = StandardScaler()
X_train =scaler.fit_transform(X_train)
X_test=scaler.transform(X_test)
```

5.To, fit and transform train and test dataset.

6.Initialize Knn Classifier.

```
knn = KNeighborsClassifier(n_neighbors=5)
y_pred = knn.fit(X_train, Y_train).predict(X_test)
```

7.To calculate the accuracy of the model

```
acc = accuracy_score(Y_test,y_pred)
```

```
print('\nAccuracy for KNN is :')
print(acc)
```

The accuracy is 89.52%.

V1.RESULT

Based on the dataset provided, the movie can be assessed to be a success. The knn algorithm produces accurate and correct results. The dataset contains information on the film. This is classified, and the model correctly predicted the film's success.

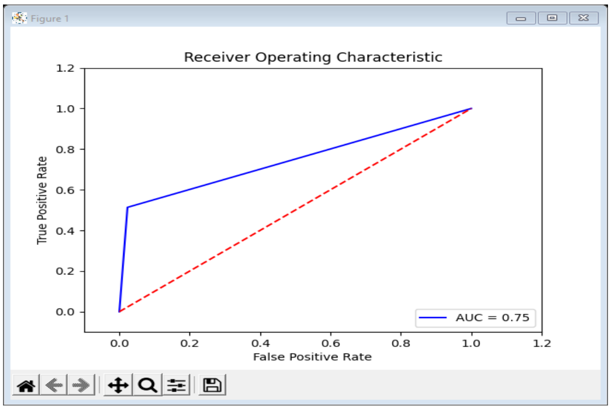

Fig 1:Model Evaluation

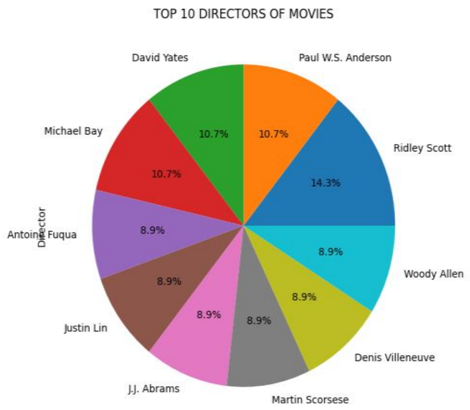

Fig 2:Top 10 movies rating

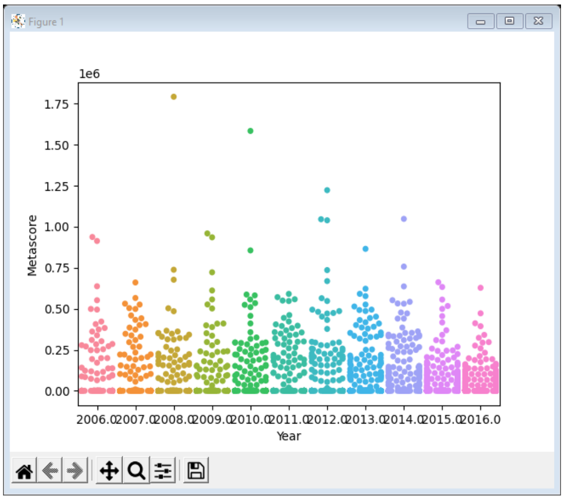

Fig 3: Rating analysis-medium rated movies

V11.CONCLUSION

Using the knn algorithm, this study proposes a method for predicting movie box office performance. This method is the most effective for predicting the success of a film. The success of a film is not only determined by these aspects of cinema. The amount of people who watch a movie is crucial to its success.

The findings suggest that box office success may be predicted with reasonable accuracy by evaluating movie emotion.

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